

Heterogeneous Treatment Effect of Red Blood Cell Transfusion in Aneurysmal Subarachnoid Hemorrhage: A Causal Machine Learning Analysis Using MIMIC-IV and eICU Databases

Study Type: Multicenter, retrospective, dual-cohort longitudinal observational study / causal machine learning-based precision medicine study

Target Journals: *Intensive Care Medicine, Critical Care, Neurocritical Care, Critical Care Medicine*

1. Background and Rationale

1.1 Clinical Background

Aneurysmal subarachnoid hemorrhage (aSAH) is an acute, highly fatal neurocritical illness predominantly affecting young and middle-aged adults, characterized by high mortality, severe disability, and substantial healthcare burden. Upon ICU admission, patients frequently develop early anemia driven by multiple interrelated mechanisms, including acute stress-induced hematopoietic suppression, recurrent minor intracranial hemorrhage, surgical trauma, and hemodilution secondary to aggressive fluid resuscitation. Anemia may further deteriorate during hospitalization due to hospital-acquired infection, systemic inflammation, and cerebral vasospasm.

Current clinical guidelines adopt a uniform hemoglobin threshold-based transfusion strategy for aSAH patients, conventionally using 8 g/dL as the transfusion trigger, though high-quality evidence specifically valid for aSAH populations remains scarce. The landmark 2024 SAHARA randomized controlled trial (RCT) published in *The New England Journal of Medicine* fundamentally updated clinical understanding: in the overall aSAH population, a liberal transfusion strategy (transfusion for Hb < 10 g/dL) yielded no improvement in 6-month neurological outcomes compared with a restrictive strategy (transfusion for Hb < 7 g/dL), presenting a neutral average treatment effect (ATE).

This authoritative negative result has sparked substantial clinical controversy. In real-world practice, individual patients exhibit highly variable prognostic responses to transfusion, with some benefiting from liberal transfusion and others experiencing exacerbated clinical deterioration. Accordingly, the neutral overall outcome of the SAHARA trial does not indicate that transfusion is clinically ineffective; instead, it suggests that **heterogeneous individual treatment effects are masked by the population-averaged effect**, leaving substantial individualized treatment heterogeneity unexplored.

1.2 Limitations of Existing Evidence

Existing aSAH transfusion studies, including the pivotal SAHARA RCT, present notable methodological limitations. First, most studies only evaluate population-averaged treatment effects while ignoring inter-individual variability in cerebral perfusion, microcirculatory function, tissue oxygenation, shock severity, and systemic inflammation, failing to distinguish transfusion-beneficial, ineffective, and harmful subgroups.

Second, the prespecified subgroup analyses in the SAHARA trial rely solely on simple stratification by single variables such as age, sex, baseline hemoglobin, and WFNS grade. These conventional univariate subgroup explorations cannot integrate multidimensional baseline phenotypic profiles, thus unable to characterize deep-level heterogeneous treatment effects (HTE) or elucidate the core mechanisms underlying variable patient outcomes.

Finally, previous real-world studies suffer from two critical methodological flaws: narrow enrollment criteria with limited population representativeness, restricting analyses to only early anemic patients and preventing direct comparison with the full-spectrum SAHARA RCT cohort; and inadequate control of causal biases induced by **time-varying transfusion treatment and dynamic physiological baseline**, lacking standardized frameworks for evaluating dynamic transfusion strategies. To date, no clinically actionable individualized transfusion decision-making system has been established for aSAH patients.

1.3 Research Gaps and Innovations

Synthesizing the findings and limitations of landmark RCTs and prior real-world studies, four critical research gaps remain in the field of precision transfusion for aSAH:

1. No studies have systematically quantified individualized heterogeneous transfusion effects based on multidimensional pre-treatment dynamic physiological phenotypes to disentangle hidden subgroup effects underlying the neutral SAHARA trial results.
2. No real-world full-cohort studies have strictly matched the enrollment criteria and intervention definitions of the SAHARA trial to enable external validation and mechanistic interpretation of RCT findings.
3. Prior studies have failed to address causal biases introduced by real-world time-varying treatment crossover and dynamic baseline shifts, lacking standardized evaluation frameworks

for transfusion strategy adherence.

4. No causal machine learning-based individualized transfusion stratification models integrating both statistical significance and minimal clinically important difference have been developed to replace conventional one-size-fits-all threshold strategies and guide bedside precision transfusion decisions.

Core Innovations: This study strictly replicates the full enrollment criteria and transfusion threshold definitions of the SAHARA trial. By innovatively incorporating **transfusion strategy adherence screening, dynamic baseline time-window calibration, and minimal clinically important difference (MID)-based stratification**, we leverage dual-center critical care databases and state-of-the-art causal machine learning approaches to systematically characterize transfusion-related heterogeneous treatment effects in a real-world full-spectrum aSAH cohort. This design mitigates dynamic treatment biases, uncovers benefit and harm subgroups masked by population-averaged effects, and provides high-quality evidence for interpreting the SAHARA negative results and establishing individualized transfusion strategies.

1.4 Scientific Hypotheses

Transfusion outcomes in aSAH patients are not uniform but individualized and determined by dynamic pre-transfusion physiological phenotypes. The neutral overall outcome of the SAHARA trial results from the mutual offset of treatment effects across beneficial, ineffective, and harmful subgroups. Based on pre-transfusion physiological profiles and standardized MID criteria, aSAH patients can be categorized into three clinically meaningful transfusion response subtypes:

- 1. Transfusion-beneficial subtype:** Patients with baseline cerebral hypoperfusion, hyperlactatemia, elevated shock index, and insufficient tissue oxygenation derive substantial clinical benefits from liberal transfusion, with significantly reduced individual in-hospital mortality risk exceeding the minimal clinically important threshold.
- 2. Transfusion-ineffective subtype:** Hemodynamically stable patients with intact tissue oxygenation and microcirculatory function exhibit no statistically or clinically meaningful outcome differences between liberal and restrictive transfusion strategies.
- 3. Transfusion-harmful subtype:** In stable patients without tissue hypoxia, liberal transfusion increases circulatory volume overload, elevates risks of transfusion-related acute lung injury, systemic inflammatory response, and microcirculatory disturbance, and consequently increases individual mortality risk.

Prespecified subgroup hypothesis: aSAH patients with early anemia within 48 hours of ICU admission and timely transfusion intervention within a short time window demonstrate more pronounced transfusion response heterogeneity and a higher proportion of liberal transfusion beneficiaries, representing the critical time window for precision clinical intervention.

1.5 Primary Study Objectives

Based on the MIMIC-IV and eICU multicenter critical care databases, this study strictly adheres to the SAHARA RCT enrollment framework and transfusion threshold definitions. By implementing standardized transfusion strategy adherence filtering and dynamic baseline time-window calibration to mitigate real-world causal biases, we apply a double machine learning causal forest model to estimate individualized conditional average treatment effects (CATE) of red blood cell transfusion in aSAH patients. Integrating statistical significance and MID-based clinical judgment, we aim to systematically verify transfusion-related heterogeneous treatment effects, identify core physiological modifiers driving differential transfusion responses, and establish a cross-center validated, clinically applicable individualized transfusion stratification system. Prespecified dynamic subgroup analyses are performed to elucidate the intrinsic mechanisms underlying the neutral overall findings of the SAHARA trial and provide high-quality real-world evidence for bedside individualized transfusion decision-making in neurocritical care.

2. Study Design and Population

2.1 Study Design

This is a multicenter, retrospective, dual-cohort longitudinal observational study comprising a model training cohort and an independent external validation cohort. All data processing, analytical pipelines, and manuscript reporting comply with the STROBE guidelines for observational epidemiological studies and standardized causal inference reporting specifications. A formal causal directed acyclic graph (DAG) is constructed to explicitly delineate confounding pathways and causal structures.

Training Cohort: MIMIC-IV v2.2 database, used for model construction, individualized CATE estimation, HTE testing, clinical subtype identification, and subgroup analyses.

External Validation Cohort: eICU Collaborative Research Database, used for verifying model stability, cross-center generalizability, and reproducibility of study conclusions.

Core Benchmark Design: The study fully adopts the SAHARA RCT full-population enrollment logic without baseline hemoglobin restrictions and strictly replicates the 7 g/dL and 10 g/dL transfusion threshold definitions. A novel **full-course transfusion strategy adherence screening criterion** is incorporated to resolve real-world dynamic treatment crossover bias, enabling direct comparison, validation, and mechanistic extension of landmark RCT results using real-world clinical data.

2.2 Data Sources

Database	Description	Study Purpose

MIMIC-IV	US tertiary hospital ICU data (2008–2019)	Model construction, CATE estimation, HTE verification, clinical subtype mining, subgroup analysis
eICU	US multicenter ICU data (2014–2015)	Model stability validation, cross-center generalizability testing, result reproducibility verification

Both databases are publicly available via PhysioNet with de-identified patient information. Secondary analyses of these databases have obtained institutional ethical approval with waived informed consent, ensuring compliant clinical data mining.

2.3 Inclusion Criteria (SAHARA RCT Alignment + Dynamic Strategy Optimization)

Patients must satisfy **all** of the following criteria for enrollment, with no baseline hemoglobin restrictions to ensure full-spectrum aSAH population coverage. Novel adherence criteria are implemented to eliminate dynamic treatment bias:

1. Age ≥ 18 years (adult patients only, minors excluded).
2. Definite diagnosis of non-traumatic aneurysmal subarachnoid hemorrhage via standardized ICD codes: ICD-9 (430), ICD-10 (I60.x); traumatic subarachnoid hemorrhage is strictly excluded.
3. ICU length of stay ≥ 24 hours, ensuring sufficient observation windows for anemia progression and transfusion assessment.
4. Availability of complete pre-transfusion dynamic physiological and laboratory variables to support covariate adjustment and causal machine learning modeling.
5. **Full transfusion strategy adherence criterion (core updated rule):** All in-hospital transfusion behaviors strictly conform to a single predefined strategy without crossover. Liberal group: transfusions only triggered by $Hb < 10$ g/dL, no non-indicated transfusions within 7–10 g/dL or at $Hb \geq 10$ g/dL. Restrictive group: transfusions only triggered by $Hb < 7$ g/dL, no non-indicated transfusions at $Hb \geq 7$ g/dL, completely eliminating time-varying treatment confounding.

Prespecified Dynamically Calibrated Subgroup Stratification:

1. **Early anemia core subgroup:** Minimum $Hb < 10$ g/dL within 48 hours of ICU admission with initial transfusion administered within 24 hours after anemia detection, reflecting real-world early clinical intervention scenarios.

2. **Severe anemia subgroup:** Dynamic pre-transfusion baseline Hb < 8 g/dL, exploring optimal transfusion strategies for severely anemic patients.

3. **SAHARA high-risk phenotype subgroup:** Enrolling patients with severe neurological injury and intraventricular hemorrhage to verify transfusion benefit heterogeneity in high-risk populations.

2.4 Exclusion Criteria

Patients meeting **any** of the following conditions are excluded to minimize confounding and data interference:

1. ICU length of stay < 24 hours with insufficient observation and data acquisition windows.
2. Severe multiple trauma, active massive hemorrhage, or massive transfusion (≥ 5 units RBCs) within the first 24 hours of admission, with drastically disturbed baseline status precluding objective transfusion effect evaluation.
3. >30% missing values in core demographic, physiological, and laboratory baseline variables, failing causal modeling data quality standards.
4. Admission with advanced malignancy, end-stage multiple organ failure, or confirmed brain death with extremely poor baseline prognosis interfering with transfusion effect assessment.
5. Patients with crossover transfusion behaviors or off-protocol transfusions that cannot be strictly assigned to a single strategy group (excluded from primary analysis and incorporated only in sensitivity analyses).

3. Variable Definition and Data Preprocessing

3.1 Core Variable Definition and Dynamic Time-Window Specification (Core Revision)

Dynamic causal temporal standardization (complete resolution of time-lag bias):

Conventional static admission baseline (T_0) is abandoned. The time point of **initial red blood cell transfusion** is defined as the dynamic T_0 . All demographic, disease severity, vital sign, and laboratory covariates are extracted from the **most recent valid measurement within 48 hours prior to initial transfusion**, precisely capturing the immediate pre-intervention physiological state and resolving the fundamental bias of static admission baselines failing to reflect delayed pre-transfusion conditions. For patients receiving immediate admission transfusions, the earliest measurements within 24 hours of admission are adopted to ensure strict causal temporality (baseline prior to exposure). All variables follow a rigorous "baseline $\rightarrow$ intervention $\rightarrow$ outcome" causal sequence to avoid reverse causality and temporal confounding.

Due to structural limitations of public databases, classical aSAH grading metrics including Hunt-Hess grade and Fisher grade, as well as invasive intracranial pressure and brain tissue oxygenation monitoring data, are unavailable. This study adopts well-validated surrogate indicators including GCS score, pupillary response, and mean arterial pressure to assess aSAH disease severity. **Bias quantification statement:** This surrogate system may slightly attenuate disease stratification accuracy and marginally reduce the detectable magnitude of heterogeneous treatment effects without generating false-positive findings, with explicit bias direction clarified in study limitations.

Variable Category	Specific Variables	MIMIC-IV Data Source
Demographics	Age, sex, race, admission type	patients, admissions
Disease Severity	Admission SOFA score, initial GCS score, abnormal pupillary response	chartevents, derived calculation
Anemia-Related Indicators	48h pre-transfusion baseline Hb, minimum Hb within 48h of admission, Hb decline rate, anemia onset timestamp	labevents (ID: 51222)
Pre-transfusion Dynamic Vital Signs	48h pre-T ₀ mean arterial pressure, heart rate, oxygen saturation, fraction of inspired oxygen	chartevents (ID: 220181, 220045, 220277, 223835)
Pre-transfusion Dynamic Laboratory Indicators	48h pre-T ₀ peak lactate, base excess, platelet count, serum creatinine	labevents (ID: 50813, 50808, 51265, 50912)
Derived Physiological Indicators	Shock index (heart rate/systolic blood pressure), oxygenation index (PaO ₂ /FiO ₂)	Standardized manual calculation
Exposure Variable	Initial transfusion strategy (strict SAHARA threshold matching + full adherence verification)	inputevents (ID: 225795, 226368)
Outcome Variables	In-hospital mortality (primary outcome); ICU	admissions, icustays, chartevents

	length of stay, mechanical ventilation duration, acute kidney injury incidence (secondary outcomes)	
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3.2 Standardized Transfusion Grouping Definition (1:1 SAHARA Replication + Adherence Optimization)

Transfusion grouping definitions strictly replicate the core framework of the SAHARA trial with additional full-course adherence validation to eliminate real-world dynamic treatment crossover bias, enabling direct and unbiased comparison between real-world evidence and RCT results:

- 1. **Liberal Transfusion Group (T=1):** Red blood cell transfusion initiated only when Hb < 10 g/dL throughout ICU hospitalization; no non-indicated transfusions within 7–10 g/dL or at Hb ≥ 10 g/dL, with full adherence to the liberal strategy.
- 2. **Restrictive Transfusion Group (T=0):** Red blood cell transfusion initiated only when Hb < 7 g/dL. Patients with persistent Hb ≥ 7 g/dL without transfusion are uniformly assigned to the restrictive strategy group, with no non-indicated transfusions within 7–10 g/dL.
- 3. **Borderline Uncertain Group (sensitivity analysis only):** Patients receiving non-essential transfusions at 7–10 g/dL or with strategy crossover that cannot be assigned to either standard group are excluded from primary analysis and retained only for robustness validation.

Exposure Temporal Definition: The time point of initial compliant transfusion is defined as the dynamic exposure onset, simulating the intention-to-treat analytical logic of RCTs. All covariates are strictly extracted prior to exposure to eliminate reverse causality and time-varying confounding. Secondary and subsequent transfusions during hospitalization are evaluated exclusively in sensitivity analyses.

3.3 Data Preprocessing and Standardization (Detailed Imputation Specification)

- 1. **Cross-database standardization:** Uniform variable naming, unit conversion, and timestamp calibration for MIMIC-IV and eICU datasets with standardized dynamic baseline extraction logic to eliminate systemic inter-database discrepancies and ensure cohort consistency.
- 2. **Missing value handling (methodological refinement):** Variables with overall missing rate >30% are directly excluded. Mild-to-moderate missing data are imputed via multiple imputation by chained equations (MICE) with predictive mean matching.**Imputation model specification:** The imputation framework incorporates all covariates, outcome variables, and temporal markers. Passive imputation is applied for derived variables (shock index, oxygenation index) to

prevent imputation bias and overfitting while preserving original data distribution characteristics.

3. Outlier cleaning: Physiologically implausible extreme values are truncated based on clinical reference ranges (MAP: 30–200 mmHg, lactate: 0–20 mmol/L) to exclude device artifacts and entry errors.

4. Data standardization: All continuous dynamic baseline covariates are standardized to zero mean and unit variance to optimize fitting efficiency and stability for causal forest and XGBoost machine learning models.

4. Causal Machine Learning Analytical Pipeline (Core Methodological Revision)

4.1 Overall Analytical Workflow (Redundant IPTW Step Removed)

SQL full-cohort extraction → dynamic time-window baseline acquisition → cross-database calibration → data cleaning and outlier exclusion → multiple imputation completion → baseline balance assessment (IPTW visualization only) → double machine learning causal forest modeling (built-in orthogonalization adjustment) → individualized CATE estimation → multi-dimensional HTE statistical testing → SHAP interpretability analysis → MID-based clinical subtype stratification → prespecified dynamic subgroup analysis → CATE distribution visualization → eICU external cohort validation → multi-layered sensitivity analysis → result visualization

4.2 Causal Estimand and Model Specification (Dual-Correction Bias Resolution)

The primary analytical target is the individualized conditional average treatment effect (CATE) of liberal versus restrictive red blood cell transfusion on in-hospital mortality in aSAH patients, defined as:

$$\tau(x) = E[Y(1) - Y(0) \mid X = x]$$

Where $Y(1)$ denotes the potential outcome under liberal transfusion, $Y(0)$ denotes the potential outcome under restrictive transfusion, and X represents the full set of 48h pre-transfusion dynamic baseline covariates.

Core methodological optimization (resolving dual-correction bias): Pre-modeling IPTW weighting is eliminated. High-dimensional confounding, propensity score imbalance, and nonlinear interaction effects are exclusively adjusted via the **built-in double machine learning orthogonalization algorithm of CausalForestDML**, avoiding standard error bias caused by duplicate correction. **IPTW is only used for baseline characteristic balance visualization (SMD < 0.1)** and not incorporated into model training or effect estimation.

Model parameter configuration: The double machine learning causal forest (CausalForestDML) is constructed via the Python econml library, with gradient boosting decision trees adopted as the outcome and nuisance models. Five-fold cross-fitting is applied to eliminate overfitting bias. Fixed hyperparameters: 100 decision trees, minimum leaf size of 10, maximum tree depth of 5, balancing model fitting accuracy and generalizability.

4.3 Heterogeneous Treatment Effect (HTE) Statistical Testing

Four complementary authoritative methods are integrated to robustly verify transfusion-related heterogeneous treatment effects in the overall cohort and subgroups:

Testing Method	Method Description	HTE Positive Criteria
Likelihood Ratio (LR) Test	Comparison of nested logistic regression models with/without CATE-by-treatment interaction terms	$P < 0.05$
CATE 95% Confidence Interval Analysis	Proportion of patients with non-zero-crossing CATE confidence intervals	Proportion $> 10\%$ indicates clinically significant heterogeneity
GATES Trend Test	Monotonic trend testing of treatment effects across CATE quintile strata	Spearman $\rho > 0$ and $P < 0.05$
SHAP Feature Importance Analysis	Quantification of core physiological modifiers driving individualized transfusion heterogeneity	Auxiliary mechanistic interpretation (non-statistical test)

4.4 Explainable AI and Clinical Subtype Stratification (MID Standard Incorporated)

An XGBoost surrogate model is fitted to individual CATE values, with SHAP explainability analysis quantifying the marginal contribution of each dynamic baseline variable to transfusion response heterogeneity to resolve the black-box limitation of machine learning models. **Novel minimal clinically important difference (MID) criteria:** Adopting critical care consensus thresholds, an **absolute risk difference $\geq 5\%$** is defined as clinically meaningful to eliminate misleading statistical significance of trivial effects. Integrating CATE values, 95% confidence intervals, and MID thresholds, all patients are standardized into three clinically valid subtypes:

1. Transfusion-beneficial subtype: $CATE < -5\%$ (liberal transfusion reduces mortality risk by $\geq 5\%$ absolutely) with 95%CI upper bound < 0 , indicating both statistical significance and

genuine clinical benefit.

2. Transfusion-ineffective subtype: CATE 95% confidence interval includes zero or absolute risk difference $< 5\%$, with no statistically significant or clinically meaningful prognostic difference.

3. Transfusion-harmful subtype: CATE $> 5\%$ with 95%CI lower bound > 0 (liberal transfusion increases mortality risk by $\geq 5\%$ absolutely), confirming definite clinical harm.

Novel visual analysis: A CATE distribution histogram is plotted for the full cohort to intuitively demonstrate subtype separation and validate the objective existence of treatment heterogeneity. Relative risk (RR) and number needed to treat (NNT) are additionally reported for each subtype to refine effect quantification.

4.5 Prespecified Dynamic Subgroup Analysis

To balance full-spectrum SAHARA benchmarking and early transfusion-focused innovation, three dynamically calibrated key subgroups are analyzed with complete HTE and CATE pipelines:

- 1. Early anemia core subgroup:** Patients with Hb < 10 g/dL within 48 hours of admission and transfusion initiated within 24 hours of anemia detection, focusing on early precision intervention scenarios.
- 2. Severe anemia subgroup:** Dynamic pre-transfusion baseline Hb < 8 g/dL, exploring optimal transfusion strategies for severely anemic patients.
- 3. SAHARA high-risk phenotype subgroup:** Patients with severe neurological injury and intraventricular hemorrhage to verify transfusion benefit heterogeneity in high-risk populations.

4.6 External Validation Framework

The full model, dynamic baseline extraction rules, and subtype stratification criteria trained on the MIMIC-IV cohort are directly transferred to the independent eICU cohort without retraining for rigorous cross-center external validation. Validation metrics include CATE quintile prognostic calibration, cross-cohort CATE Spearman correlation, Log-rank consistency of subtype survival curves, and subtype classification accuracy, comprehensively verifying model generalizability and result stability.

5. Outcome Measures and Statistical Analysis Plan

5.1 Primary Outcome

All-cause in-hospital mortality (short-term prognosis): All-cause mortality prior to hospital discharge, serving as the primary hard endpoint for evaluating short-term prognostic effects of transfusion strategies. The manuscript uniformly specifies "short-term prognosis" in the title, abstract, and full text to avoid misleading interpretation of long-term outcomes.

5.2 Secondary Exploratory Outcomes

- 1. Total ICU length of stay
- 2. Invasive mechanical ventilation duration
- 3. Incidence of acute kidney injury defined by the 2012 KDIGO criteria

Fine-Gray competing risk models are applied for time-dependent outcomes to adjust for mortality competing bias and enhance analytical accuracy.

5.3 Comprehensive Statistical Analysis Plan (Power Calibration Included)

- 1. **Baseline balance evaluation:** Standardized mean difference (SMD) is used to assess post-IPTW covariate balance, with $SMD < 0.1$ indicating adequate confounding control.
- 2. **Integrated HTE validation:** Heterogeneity is comprehensively verified via likelihood ratio testing, GATES trend analysis, CATE interval evaluation, and CATE distribution visualization.
- 3. **Post-hoc power assessment and limitation statement:** The combined dual-cohort sample size (3000–4000 patients) ensures robust detection of moderate-to-large heterogeneous effects ($\geq 15\%$ absolute risk difference). The early anemia subgroup (approximately 1500 patients) achieves $>80\%$ statistical power for detecting risk differences $\geq 20\%$. **Explicit power limitation:** The study is underpowered to detect small-magnitude heterogeneity (5%–10% absolute risk difference), with potential omission of subtle true effects. Multiple-imputation interaction power simulation will be performed for further optimization.
- 4. **Multi-layered sensitivity analysis (temporal sensitivity added):** Complete case analysis for missing data robustness; alternative grouping analysis incorporating borderline transfusion patients; negative control outcome testing to exclude spurious heterogeneity; permutation testing for result reliability; E-value analysis to quantify unmeasured confounding; and novel subsequent transfusion temporal sensitivity analysis to evaluate the impact of secondary transfusions on primary outcomes.
- 5. **Causal visualization supplementation:** A standardized causal directed acyclic graph (DAG) is constructed to explicitly delineate causal pathways of exposure, outcome, confounders, time-varying covariates, and colliders, standardizing causal inference logic.

All statistical tests are two-tailed with a significance level of $\alpha = 0.05$. All analyses are implemented using professional Python analytical libraries.

6. Study Timeline (Revised Full Version)

Phase	Core Tasks	Estimated Duration

Phase 1	SQL full-cohort extraction, dynamic time-window baseline construction, strategy adherence screening, cross-database variable calibration	3–4 weeks
Phase 2	Data cleaning, outlier elimination, standardized multiple imputation, IPTW baseline balance analysis	2 weeks
Phase 3	Causal forest model training, hyperparameter tuning, individualized CATE batch estimation	2–3 weeks
Phase 4	Multi-dimensional HTE testing, SHAP mechanistic analysis, MID-based subtype stratification, CATE distribution visualization, DAG plotting	1–2 weeks
Phase 5	eICU external validation, dynamic subgroup in-depth analysis, full sensitivity analysis, figure and table optimization	2 weeks
Phase 6	Manuscript drafting, logical refinement, methodological supplementation, revision, and journal submission	4–6 weeks
Total Duration	Excluding journal peer review cycles	14–19 weeks

7. Innovations and Clinical Translation Value

7.1 Methodological Innovations

1. Precise benchmarking against top-tier RCTs to resolve core clinical controversies: This study achieves the first 1:1 replication of SAHARA trial full-population enrollment and

transfusion threshold definitions in real-world data. The novel strategy adherence screening eliminates dynamic treatment crossover bias, enabling direct dialogue and complementary verification between real-world evidence and RCT results to resolve clinical disputes regarding the SAHARA negative findings.

2. Dynamic baseline design eliminating temporal shift bias: Abandoning conventional static admission baselines, the adoption of 48-hour pre-transfusion dynamic physiological baselines precisely captures immediate pre-intervention patient status, resolving the fundamental temporal mismatch bias in prior studies and substantially improving causal inference accuracy.

3. Rigorous causal design eliminating statistical biases: Redundant IPTW modeling steps are removed to avoid dual correction conflicts. Confounding is exclusively adjusted via causal forest built-in double machine learning orthogonalization, separating baseline balance visualization from model training. The integration of DAG causal mapping, MID clinical criteria, and CATE distribution visualization fully complies with top-tier journal causal inference standards, balancing statistical rigor and clinical practicality.

4. Balanced research breadth and core characteristics: Full-cohort primary analysis ensures sufficient sample size and statistical power, while dynamically calibrated early anemia subgroup analysis retains the study's core clinical focus, satisfying both comprehensive academic standards and targeted clinical relevance for high-impact publication.

7.2 Clinical Significance and Translation Value

This study fundamentally overturns the generalized conclusion of "no transfusion benefit in all aSAH patients" derived from the SAHARA trial and confirms substantial individualized heterogeneity in aSAH transfusion responses, demonstrating that neutral population-averaged outcomes arise from offsetting subtype effects. By integrating **statistical significance and minimal clinical importance** dual criteria, we accurately identify genuine transfusion-beneficial, ineffective, and harmful phenotypic subgroups, replacing traditional one-size-fits-all transfusion thresholds. The findings enable the construction of a bedside actionable individualized aSAH transfusion decision system, facilitating precise early critical care intervention, reducing ineffective and harmful transfusion practices, conserving blood resources, lowering transfusion-related complication rates, optimizing aSAH critical care management, and providing high-quality real-world evidence for future neurocritical care transfusion guideline updates.

8. Study Limitations (Comprehensive Revised Specification)

1. Residual confounding in observational studies: Despite multi-layered bias control via dynamic baseline calibration, strategy adherence filtering, and double machine learning

orthogonalization, unmeasured time-varying residual confounding cannot be entirely eliminated. The causal evidence level is inferior to prospective RCTs, though cross-benchmarking with SAHARA findings and multi-dimensional sensitivity analyses substantially enhance result credibility.

2. Disease assessment limitations and quantifiable bias: Classic aSAH grading systems (Hunt-Hess, Fisher) are unavailable in public databases, with GCS and pupillary response adopted as surrogates. This slightly attenuates disease stratification precision and may marginally reduce detectable heterogeneous effect magnitudes without generating false-positive conclusions, with explicit and controllable bias direction.

3. Statistical power limitations: The study reliably detects moderate-to-large transfusion heterogeneity but lacks sufficient power for small-magnitude effects (5%–10% absolute risk difference), with potential omission of subtle heterogeneity. Multiple-imputation interaction power simulation will be applied for further optimization.

4. Real-world selection bias: Exclusion of crossover transfusion patients ensures strategy purity but slightly reduces sample representativeness, though robustness is verified via sensitivity analysis.

5. Geographic population limitations: All data originate from US cohorts, limiting generalizability to Asian and other ethnic populations, requiring further external validation in multi-ethnic cohorts.

6. Outcome dimension limitations: Long-term 6-month mRS neurological outcomes are unavailable, restricting analyses to short-term in-hospital mortality. This limitation is explicitly stated throughout the manuscript to avoid conclusion misinterpretation.

7. Unrefined dynamic temporal effects: This study focuses on individualized effects of initial transfusion strategies without fine-grained modeling of cumulative effects of repeated transfusions and transfusion duration, providing directions for future temporal mechanism research.

9. Ethical Statement

The MIMIC-IV and eICU public databases have obtained official institutional ethical approval with de-identified patient data and waived informed consent for compliant secondary clinical data analysis. This study strictly adheres to international clinical research specifications and privacy protection principles without privacy disclosure risks, requiring no additional ethical approval.

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